

Chromoplexy, and Cancer Evolution

Linking Aberration Multigraph Theory to Prostate Cancer Genomes, Chromoplexy, and Clonal Progression in Prostate Tumor Progression.

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Gradual Accumulation Model of Cancer

How we have always thought cancer develops

- Every cell in your body accumulates random mutations over time and most mutations are harmless
- But, occasionally a mutation hits a gene that controls growth. Cancer develops when enough of these mutations accumulate
- Each mutation happens independently and gives the cell a small growth advantage. Over many years and many cell divisions, cancer emerges.
- This is called the gradual accumulation model. Like a staircase, one step at a time where each step is random and independent of the others

Current Clinical Tools to Assess Prostate Cancer Risk

Clinicians currently use two main tools to assess risk:

1. **Gleason Score:** measures how abnormal tumor cells look under the microscope. They are numerical grades assigned to different cell patterns found within the same tumor or biopsy sample.
 - **Primary grade** represents the most common pattern (making up >50% of the sample)
 - **Secondary grade** represents the next-most frequent pattern (has to be less than 50%, but at least 5%, of the pattern of the total cancer observed).
 - **Tertiary grade** represents a small component of a third (generally more aggressive) pattern. Generally accounts for less than 5% of the tumor volume.
 - Ranges from 6 to 10
 - Higher score = more disorganized cells = more aggressive

Current Clinical Tools to Assess Prostate Cancer Risk

2. **Pathologic Stage:** determined by a pathologist before and after surgery to determine how far the tumor has spread.

Clinical (c)	Pathologic (p)	Severity	What it means
cT1	(N/A)	Very Low	Hidden: Not felt on exam or seen on MRI; found only by biopsy.
cT2	pT2	Low	Localized: Cancer is strictly contained within the prostate gland.
cT3a	pT3a	Moderate	Extraprostatic: Cancer has broken through the outer shell (capsule).
cT3b	pT3b	Serious	Seminal Vesicle: Cancer has spread to the nearby semen-producing glands.
cT4	pT4	Severe	Invasion: Cancer has grown into the bladder, rectum, or pelvic wall.
cN1	pN1	Advanced	Regional Nodes: Cancer found in the pelvic lymph nodes.
cM1	pM1	Systemic	Metastasis: Cancer has spread to bones, lungs, or distant organs.

Regional Metastasis (The "N" Category)

Before spreading to distant organs, prostate cancer usually moves to the **lymph nodes** in the pelvis.

- **pN0:** No cancer found in nearby lymph nodes.
- **pN1:** Cancer is found in one or more regional lymph nodes. This is considered **Stage IV** disease even if the tumor is small.

Distant Metastasis (The "M" Category)

This is the most severe stage, where cancer travels through the blood or lymph system to other parts of the body.

- **M1a:** Spread to non-regional lymph nodes (like those in the chest or neck).
- **M1b:** Spread to the **bones** (the most common site for prostate cancer metastasis).
- **M1c:** Spread to other organs like the **liver or lungs**.

The issue

PSA = Prostate Specific Antigen. A protein produced by prostate cells. After surgery it should be undetectable, near to zero. A rising PSA means cancer cells are still present somewhere.

Two patients. Identical **Gleason 7, T3a diagnosis.**
Same age. Same PSA at diagnosis. Same surgery.

- Patient A — Cured. No recurrence after 5 years.
- Patient B — PSA rises at 18 months. Cancer returns.

Current tools cannot tell these two patients apart before it happens.
Something about the underlying biology is different between these two patients and we are not measuring it.

What if the answer lies not in how the cells look, but in how their genomes broke and rejoined?

What Baca et al. Found

Punctuated Evolution of Prostate Cancer Genomes

[Sylvan C. Baca](#)^{1,2,3} · [Davide Prandi](#)⁶ · [Michael S. Lawrence](#)² · ... · [Francesca Demichelis](#) ^{6,9,18}  · [Mark A. Rubin](#)^{8,11,18} · [Levi A. Garraway](#) ^{1,2,3,4,18}  ... [Show more](#)

Summary

The analysis of exonic DNA from prostate cancers has identified recurrently mutated genes, but the spectrum of genome-wide alterations has not been profiled extensively in this disease. We sequenced the genomes of 57 prostate tumors and matched normal tissues to characterize somatic alterations and to study how they accumulate during oncogenesis and progression. By modeling the genesis of genomic rearrangements, we identified abundant DNA translocations and deletions that arise in a highly interdependent manner. This phenomenon, which we term “chromoplexy,” frequently accounts for the dysregulation of prostate cancer genes and appears to disrupt multiple cancer genes coordinately. Our modeling suggests that chromoplexy may induce considerable genomic derangement over relatively few events in prostate cancer and other neoplasms, supporting a model of punctuated cancer evolution. By characterizing the clonal hierarchy of genomic lesions in prostate tumors, we charted a path of oncogenic events along which chromoplexy may drive prostate carcinogenesis.

Baca et al. sequenced the complete genomes of 57 prostate tumors and found a

CHRONOLOGY of Oncogenic Events in Prostate Cancer Progression.

Clonality Concept

Mutations that happen EARLY are present in EVERY tumor cell and are called **clonal mutations**.

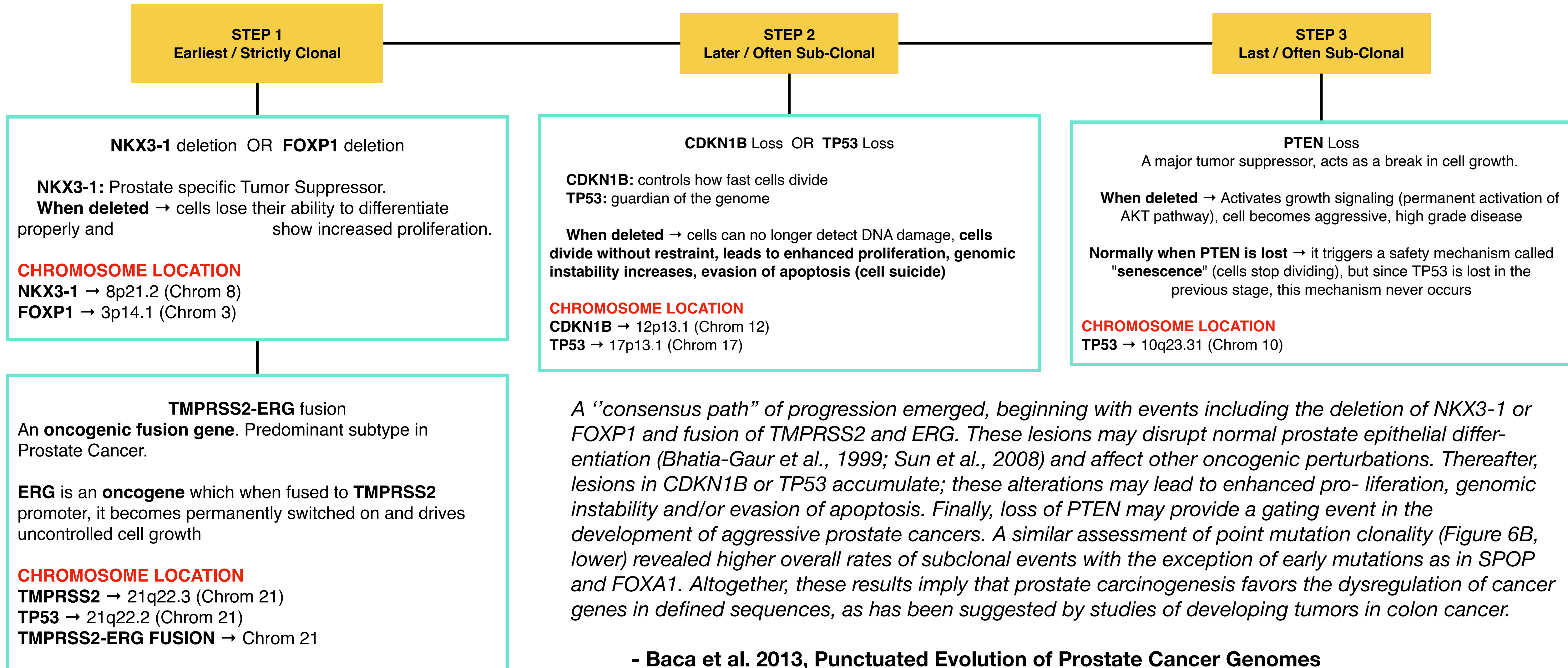
Mutations that happen LATER are present in only SOME tumor cells and are called **sub-clonal mutations**.

By measuring what fraction of tumor cells carry each mutation, Baca could reconstruct the timeline,

Clonal = happened before the tumor expanded = **EARLY**

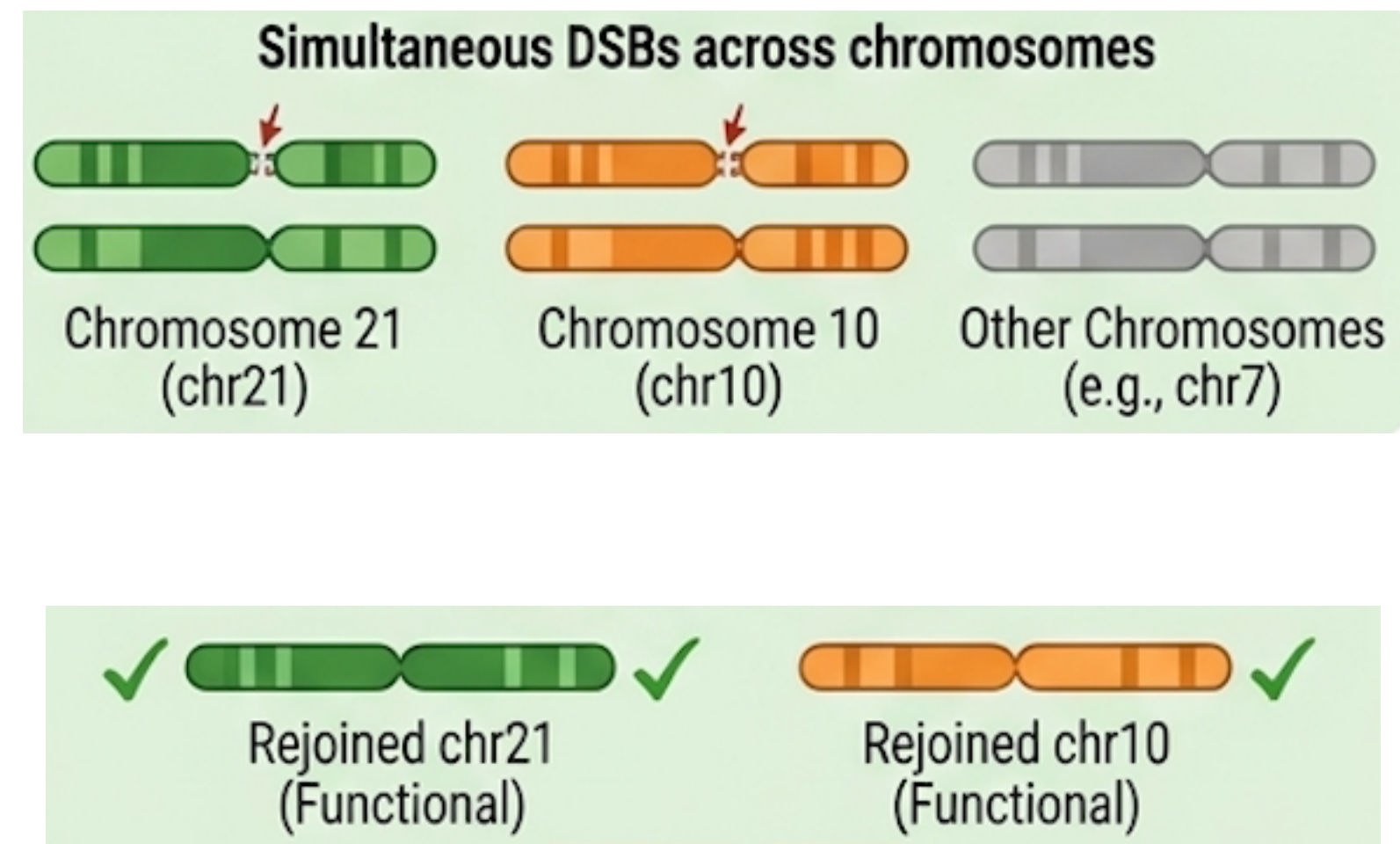
Subclonal = happened after the tumor was established = **LATE**

The Consensus Path

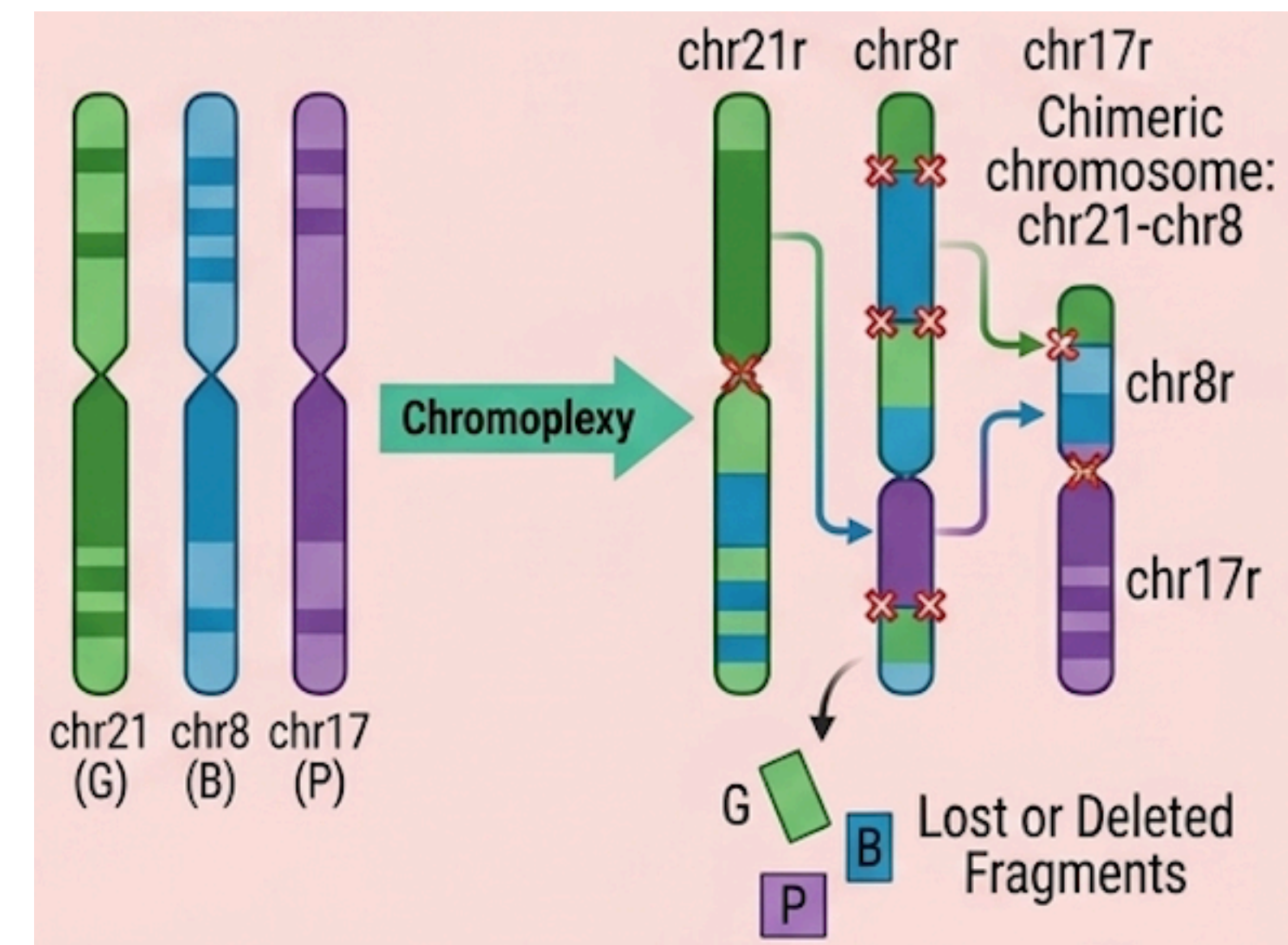


Chromoplexy: Coordinated Genome Restructuring

When multiple DNA double strand breaks occur simultaneously across different chromosomes, the broken ends can incorrectly rejoin each other in a chain.



Correct Repair



Incorrect Repair

Chromoplexy: Coordinated Genome Restructuring

Chromoplexy happens at each step of the consensus path. Baca et al. observed the following:

1. “Oncogenic ERG fusions frequently arose in the setting of chromoplexy (15 of 26 cases, 58%).”
2. “A chain of 27 rearrangements across six chromosomes include the TMPRSS2-ERG fusion (21q) as well as a disruptive rearrangement of the prostate tumor-suppressor gene SMAD4.”
3. “Several cancer genes were recurrently deleted or rearranged by chromoplexy, including PTEN (nine cases), NKX3-1 (eight cases), CDKN1B (three cases), TP53 (four cases), and RB1 (two cases). Thus, chromoplexy may conceivably influence prostate carcinogenesis by disrupting TSGs and creating oncogenic fusions.”
4. "Several chains appeared to involve strictly subclonal deletion bridges, indicating that tumors may sustain multiple rounds of chromoplexy.”
5. "Along with the observation that chromoplexy may affect both early and late genes in the consensus path (e.g., ERG and PTEN), these findings suggest that chromoplexy also occurs in tumor subclones that emerge later during cancer evolution.”

- Baca et al. 2013, Punctuated Evolution of Prostate Cancer Genomes

Baca explicitly states chromoplexy operates at both ends of the progression path, in different cells, at different times, throughout the entire life of the tumor.

Chromoplexy or Chromothripsis?

	Chromoplexy	Chromothripsis
ETS Status	ETS Fusion Positive +	ETS Fusion Negative -
CHD1 Gene	CHD1 wild-type	CHD1 deleted
Chromosomes affected	Many — up to 6 or more simultaneously	Few — typically 1 to 2
Rearrangement type	Interchromosomal — breaks rejoin across different chromosomes	Intrachromosomal — breaks rejoin within the same chromosome
AMG framework (Inferred)	Θ (large k , small b_i) — many chromosomes, few breaks each	Θ (small k , large b_i) — few chromosomes, many breaks each

AMG Framework

Every tumor's rearrangement pattern can be described as:

$$\Theta(k, (b_1, b_2, \dots, b_k))$$

Where:

k = number of chromosomes involved

b_i = number of double strand breaks on chromosome i

Examples: $\Theta(6, (1,1,1,1,1,1))$

6 chromosomes, 1 break each

→ ETS+ chromoplexy pattern

→ Must form one large single cycle C_n

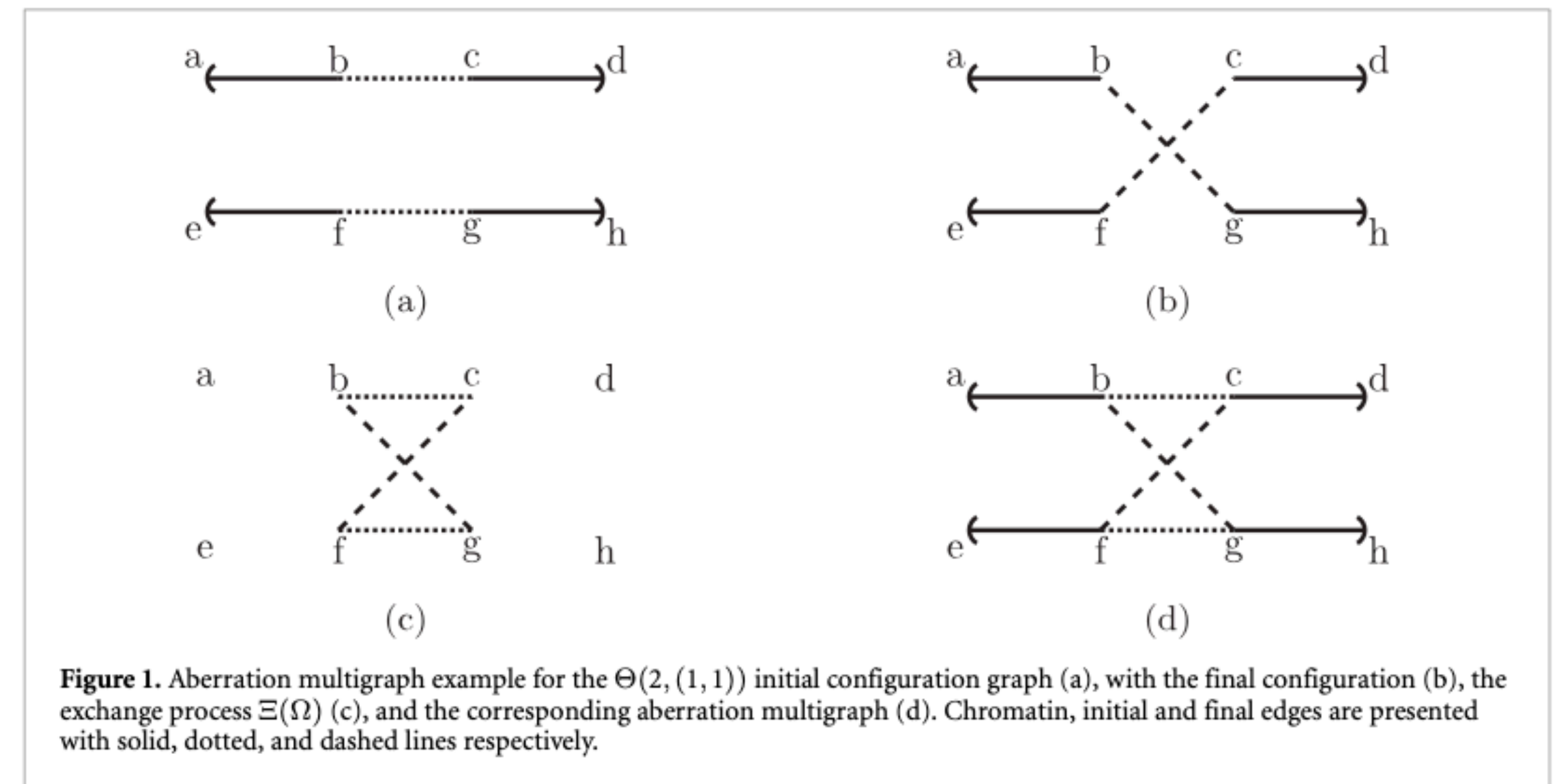
→ Lemma 14 in Sheth et al. proves this mathematically

$\Theta(1, (12))$ 1 chromosome, 12 breaks

→ CHD1del chromothripsis pattern

→ Many possible cycle structures

→ Exponentially more complexity



What questions is this project ideally trying to answer?

1. Does genomic complexity increase with **Gleason grade**, as Baca found in 57 tumors?
2. Do **ETS+** patients show higher k (more chromosomes with copy number alterations) consistent with the **chromoplexy** AMG configuration $\Theta(\text{large } k, \text{small } b_i)$?
3. Do **CHD1 - deleted** patients show higher b_i (more alterations per chromosome) consistent with the **chromothripsis** AMG configuration $\Theta(\text{small } k, \text{large } b_i)$?
4. Does **Baca's consensus progression path** correspond to **increasing rank in Sheth's poset $L(n)$** ?
That is, do patients at later progression stages occupy higher positions in the AMG complexity hierarchy?
5. Does **AMG complexity** predict **biochemical recurrence** independently of **Gleason score** and **pathologic stage**? That is does AMG cycle structure complexity predict something that Gleason score and stage do not?
6. Does the same progression sequence — **NKX3-1** → **CDKN1B/TP53** → **PTEN** — appear in 500 TCGA patients?
*(Maybe, Not Sure)